Microbiology: Fungal Pathogens of the Respiratory System

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Do.
Make.
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Reinvent the Future.

Session Objectives

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1. Identify the properties of the most common primary fungal pathogens of the respiratory tract.
2. Identify the etiology, clinical presentation, diagnosis, and treatment for each of the fungal respiratory pathogens.
3. Describe the epidemiology involved in the spread and acquisition of respiratory fungal diseases.
4. Define the pathogenesis of the fungal agents.
5. Outline the laboratory techniques used for the identification and diagnosis of these fungal pathogens.

What are Fungi?

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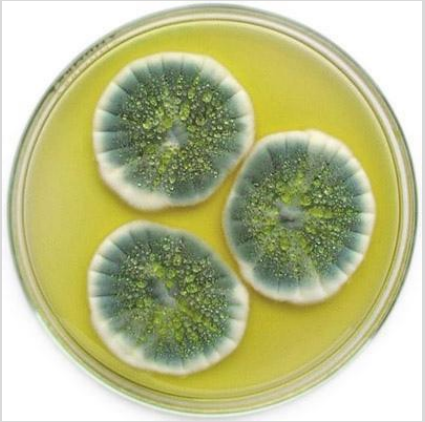
- Fungi are mostly 'free-living' eukaryotic cells
- Evolved alongside animal cells
- Non-motile, cell wall, non-photosynthetic
- Mostly very beneficial (saprophytes)

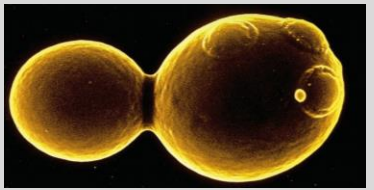
FEATURE	BACTERIA	FUNGI	VIRUSES
CELLS	YES	YES	NO
SIZE	1-5 MICRON	3-10 MICRON	0.02-0.2 MICRON
Genetic Material	DNA	DNA	DNA or RNA
Classification	PROKARYOTIC	EUKARYOTIC	N/A
RIBOSOMES	70S	80S	N/A
Membrane-bound organelles	NONE	YES	N/A
OUTER LAYER	PEPTIDOGYLCAN	CHITIN/GLUCAN	CAPSID/ENVELOPE
MOTILITY	SOME HAVE	NO	NO
REPLICATION	BINARY FISSION	BUDDING/MITOSIS	OTHER
Cell Membrane	PL/Hopanoids	PL/Ergosterol	N/A

Fungi: Basic Facts

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- Can be divided into 2 types:
  - Yeasts
  - Mold (hyphae)
- Some species are ***Dimorphic***
  - Can grow as unicellular or multicellular
  - Based on temperature
- Pathogenic fungi are unicellular
  - Must grow at 37°C for virulence
- Mostly plant pathogens (humans are accidental hosts)
- Fungi also play an important role if various industries





Systemic Fungal Infections

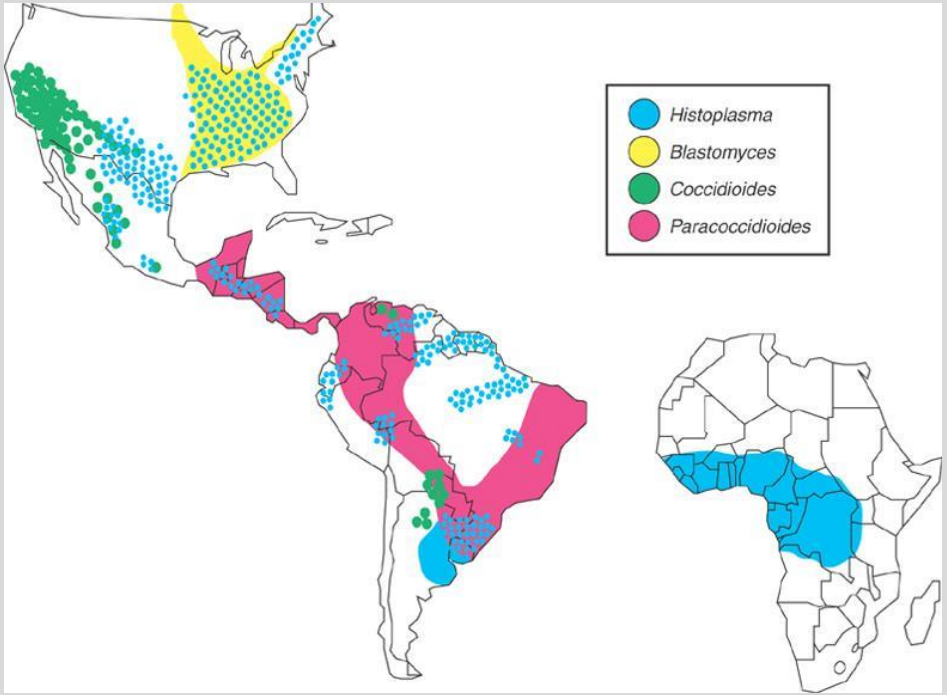
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- Uncommon
  - Most superficial
- Natural immunity is high
  - Innate and cell mediated
- Cases have increased over the past few decades
- High mortality rate, treatment is challenging
- Most are asymptomatic or self-limiting
- Most acquired through respiratory track

Systemic Fungal Disease

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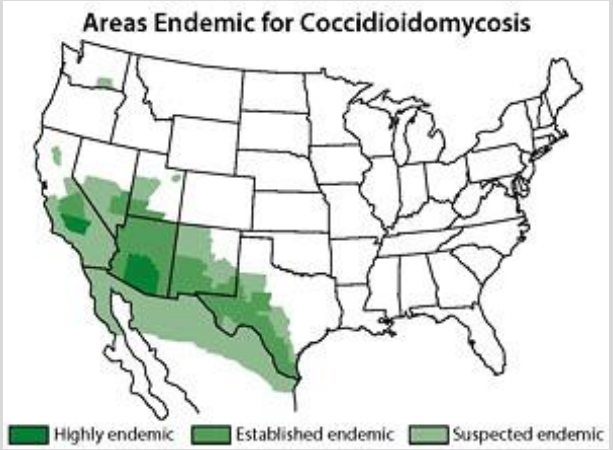
- Most often associated with four genera:
  1. *Coccidioides posadasii & C. immitis*
  2. *Histoplasma capsulatum*
  3. *Blastomyces dermatitidis*
  4. *Paracoccidioides brasiliensis**



Coccidioidomycosis

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- *Coccidioides posadasii & C. immitis* are *most virulent* of fungal pathogens
- Restricted to hot, semi-arid areas of SW USA & Mexico
- Grows in the soil
- Inhalation of arthrospores (conidia) initiate infection
- Cycles of drought/rain → dispersion
- Dimorphic
- Asymptomatic (60%)
- Highly endemic areas: 15-30% of community-acquired pneumonias
- Valley fever

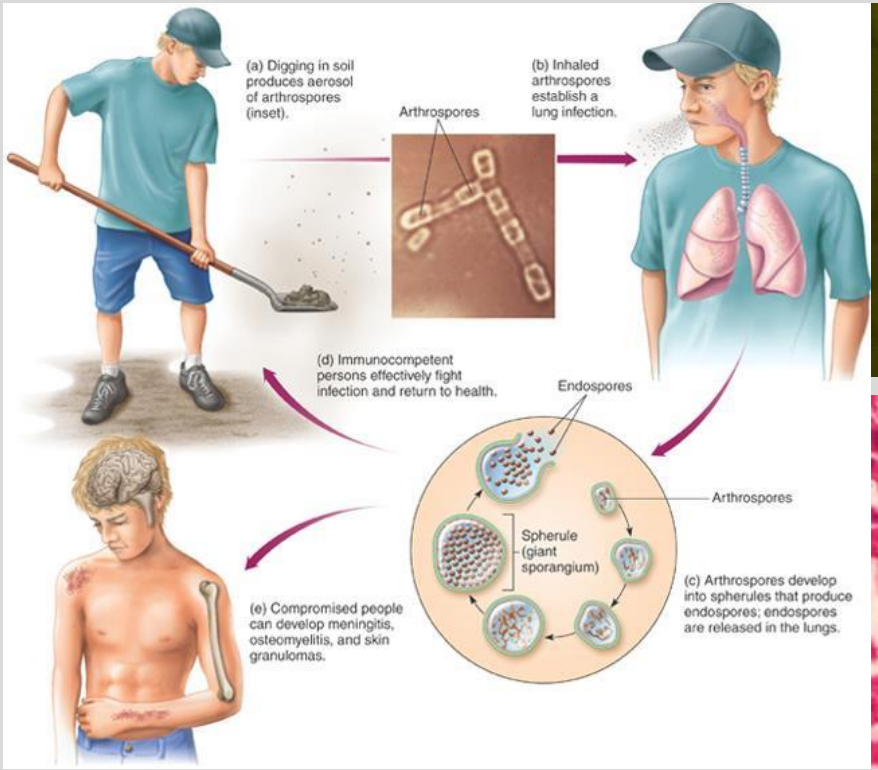


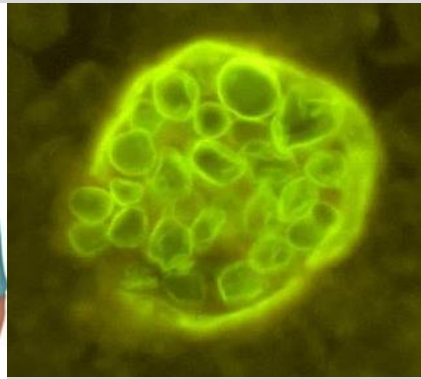


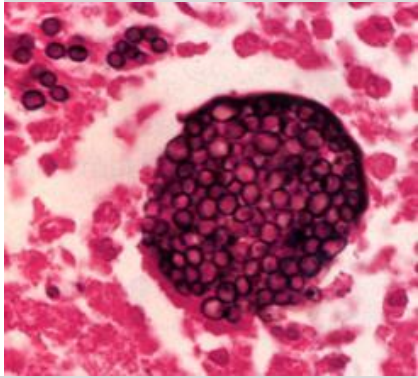
Events in *Coccidioides* infection

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- Inhalation of conidia
- Form Spherules
- Immune response: T-cell mediated (Th-1)
- Resistant to killing by phagocytes
- Symptoms
  - Flu-like
  - Erythema nodosum
- Spread: blood → skin, soft tissues, bones, joints & meninges







Coccidioidomycosis

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- **Risks Factors**
  - Ethnicity (African Americans, Hispanic & Filipinos)
  - Age: Extremes more susceptible
  - Sex: Males more susceptible
  - Pregnancy: 3rd trimester
  - Immunosuppression (HIV, transplants, etc.)
- **Diagnosis**
  - Histopathology: spherules or spores seen in sputum, exudates or tissue is sufficient
  - Culture: dangerous! (direct microscopic observation preferred)
  - Serology
- **Treatment**
  - Fluconazole or itraconazole for mild
  - Amphotericin B for disseminated
  - Meningeal coccidioidomycosis is managed with fluconazole

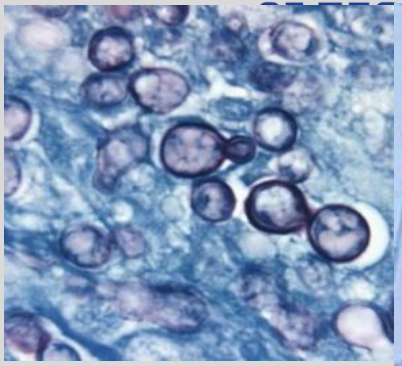




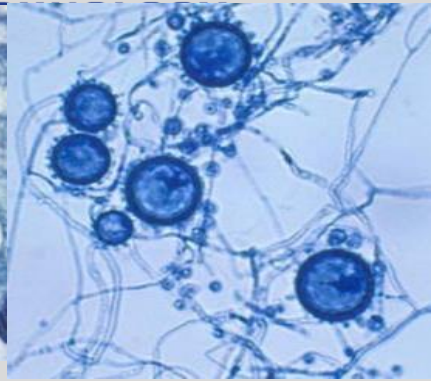
Histoplasmosis

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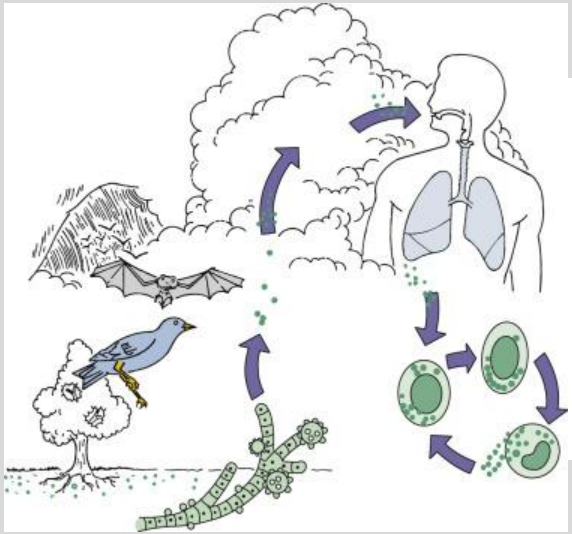
- *H. capsulatum* (US); var: *duboisii* (Africa)
- Endemic areas:
  Central & Midwestern states (most cases in Ohio Valley & Mississippi Valley)
- Dimorphic/soil saprophyte
  - Hyphae contain macro & microconidia
- Birds/bats droppings (guano)
- Cave disease

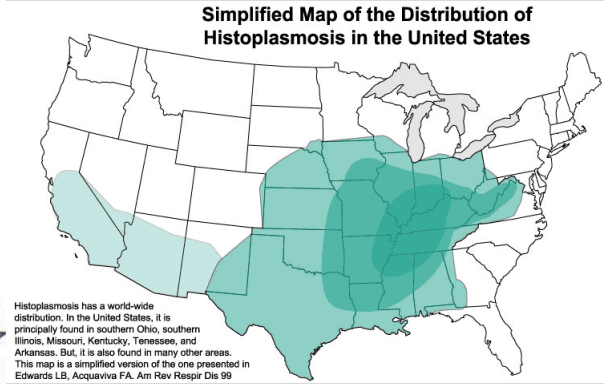


Yeast @ 37°C



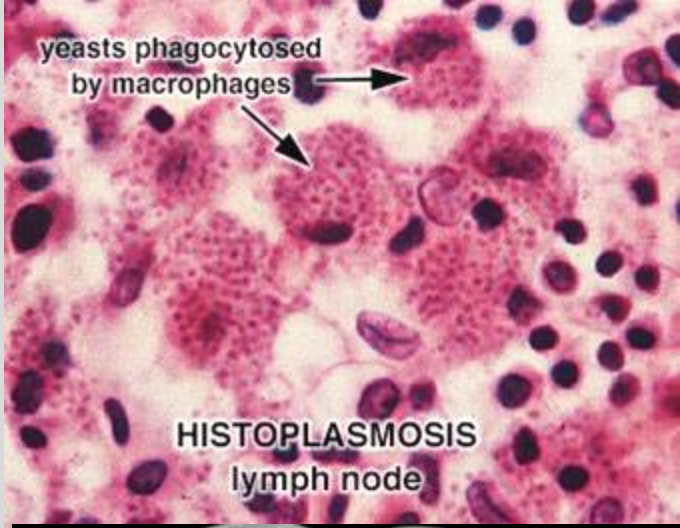
Conidia @ 25 °C





Histoplasmosis

- 90% of cases are asymptomatic
  - Rare cases show flu-like symptoms
- **Immune Response**
  - Phagocytic activity of macrophage is considered an important component of resistance
    - Survival in macrophages—elevates pH of phagosomes
    - Macrophages spread yeast
  - Cell-mediated response help clear infection
    - Activated macrophage can kill yeast cells
    - Granuloma formation (potential for reactivation)
- Disseminated histoplasmosis can involve many systems
  - Diagnosed frequently in patients with AIDS living in the central U.S.





Histoplasmosis

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- **Diagnosis**
  - Direct histology & culture of blood or bone marrow
    - Visualization of yeast cells inside macrophages
  - Serological testing for antibody or Ag in blood & urine
  - PCR

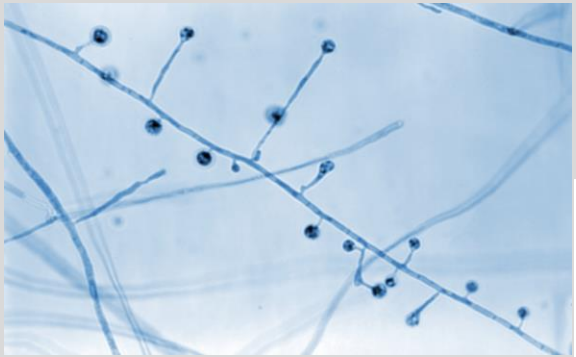
- **Treatment**
  - Itraconazole for mild or moderate disease
  - Amphotericin B for disseminated and severe pulmonary histoplasmosis

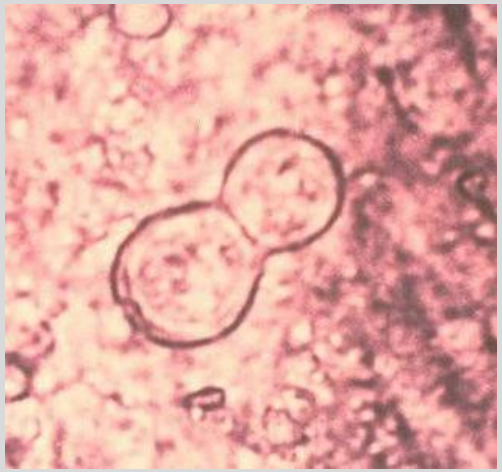


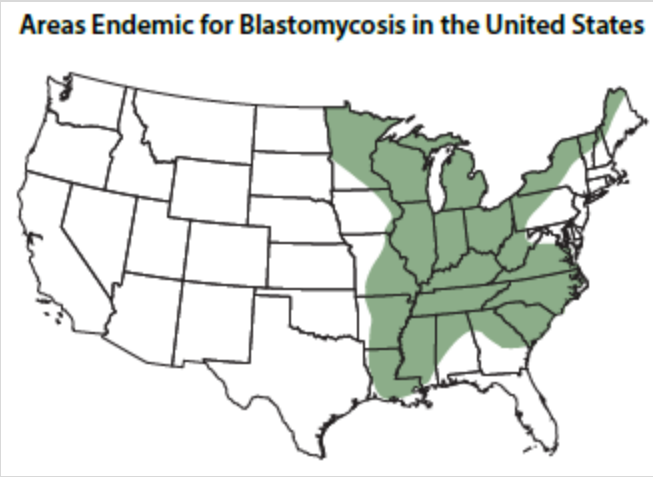
Blastomycosis

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- *Blastomyces dermatitidis*
- Habitat: moist soil, rural riverbanks
- Endemic in Northern America
- Dimorphic
    - Hyphae produce conidia (pear shaped)
    - Yeast:
        - Thick wall
        - Multinucleate
        - Broad based budding







Blastomycosis

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- **Pathogenesis and Immune response:**
  - Alveolar macrophage provide a first line of defense
    - Conidia are more sensitive to killing because yeast are too big
  - Escapes phagocytosis
  - T-cells stimulate neutrophils to kill *Blastomyces* cells
  - Pyogranulomas form
    - Granulomatous infection predominantly involves lungs
- Dissemination
  - Commonly skin, bones, GU tract

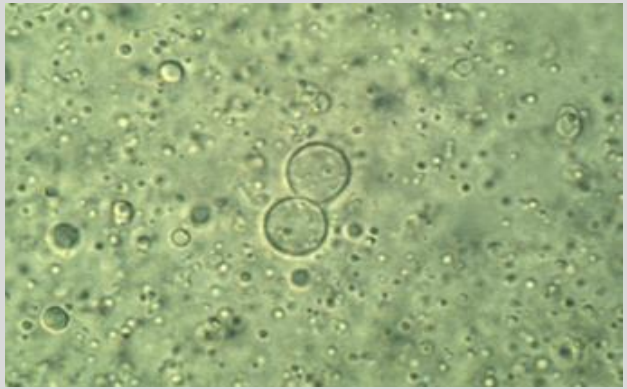


Blastomycosis

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- **Diagnosis:**
  - Microscopic detection
  - Culture
  - PCR
  - Serology is not useful
- **Treatment:**
  - Itraconazole
  - Amphotericin B

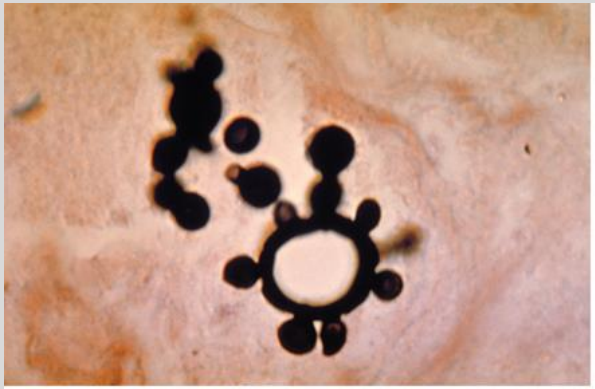
KOH preparation from sputum



Paracoccidiodomycosis

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- *Paracoccidioides brasiliensis*
- Distributed in Central and South America
- Dimorphic
    - Yeast have multiple buds (captain's wheel)
- Lung infection, common in older men (farmers, rural laborers)
- Dissemination uncommon (oral mucosa most common site)
- Treatment:
    - Itraconazole
    - Sulfonamides
    - Amphotericin B



Source: P. Chin-Hong, E. A. Joyce, M. Karandikar, M. Matloubian, L. A. Rubio, B. S. Schwartz, W. Levinson: Levinson's Review of Medical Microbiology & Immunology: A Guide to Clinical Infectious Diseases, 19th Edition: Copyright © McGraw Hill. All rights reserved.





Talaromycosis

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- *Talaromyces marneffei*
    - Former genus: *Penicillium*
- Endemic in Southeast Asia
- Disease develops in immunocompromised, especially those with HIV
- Dimorphic/soil; Opportunistic
- Yeast is elongate with central septum
- Dissemination possible
- Treatment: Amphotericin B and/or itraconazole



Summary

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- The endemic mycoses are common infections that exist within restricted geographical areas
- Primary fungal pathogens that cause systemic disease are typically thermally dimorphic
- Clinical manifestations are unique to each endemic fungus, but the lung is prominently involved
- Cell-mediated immunity is essential for successful resolution of these infections
- Impaired immunity can facilitate systemic fungal disease

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